

Programming language for cyber security

CST301

Prepared
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Methodology of working within this semester

- 1- the door will be closed after 15 minutes**
- 2- there are three online quizzes within the semester.**
- 3- the final exam will include questions are similar to the questions of our lectures**
- 4- there are project that may be antivirus, protocol, virus ... etc**
- 5- the absence will be involved in our lecture.**
- 6- our marks will be classified as following:**
60 (med term) –15 (our lecture) – 15 (the lab) – 60 (final exam)
- 7- the marks for the lecture and lab are classified between absence, activities in lecture and assignment**
- 8- the total mark is 150 and the time for exam is 2 hours**

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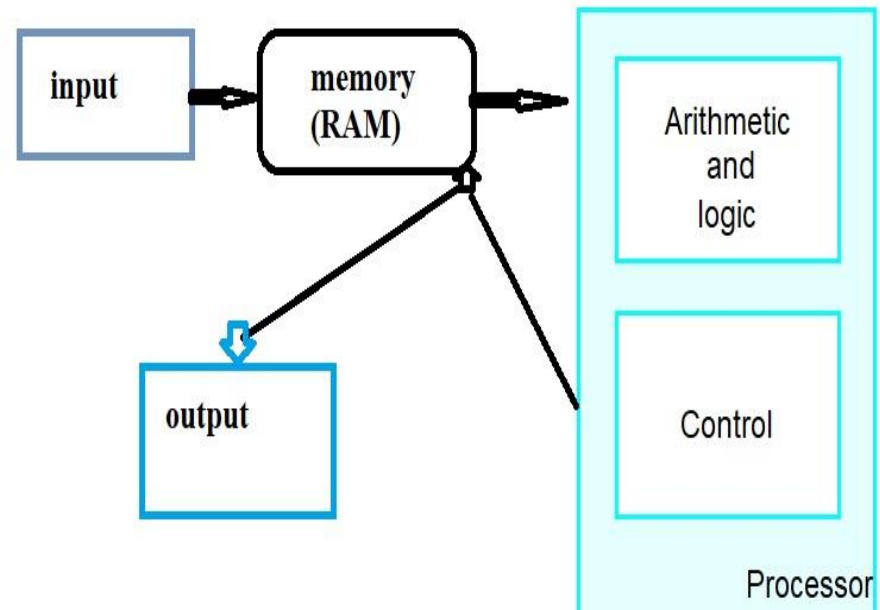
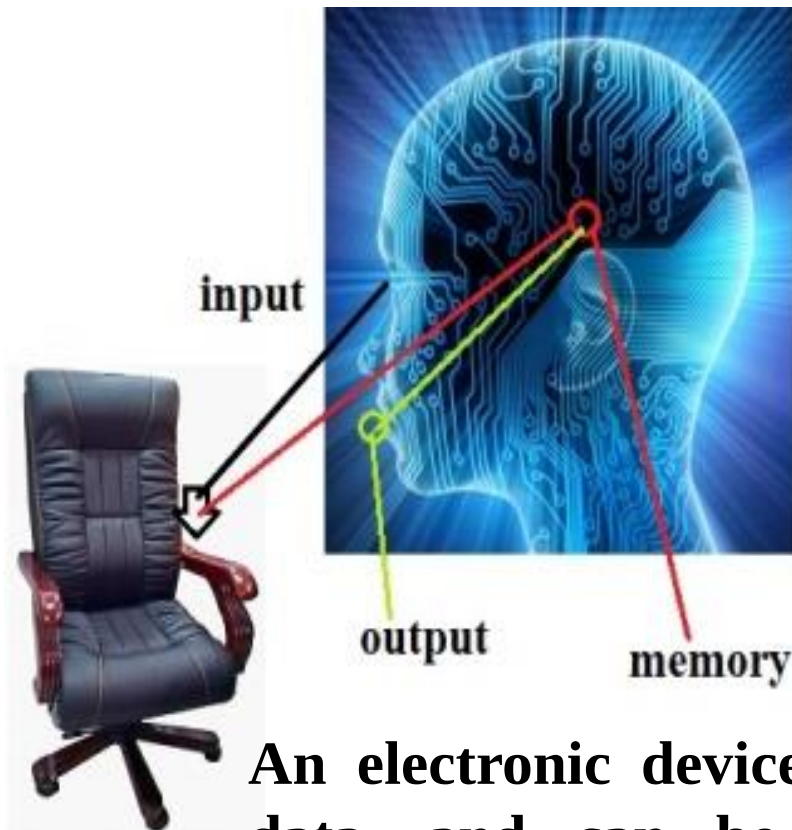
Outline

- 1- What is a Computer?**
- 2- why a computer is?**
- 3- what are types of computer?**
- 4- what is a Programming Language?**
- 5- the questions on the lecture**

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1- What is a Computer?



An electronic device that stores, retrieves, and processes data, and can be programmed with instructions. A computer is composed of hardware and software, and can exist in a variety of sizes and configurations.

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1- What is a Computer?

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Four units are found in every computer:

1. Input unit

- Gets the information from user within the input devices (keyboard, mouse, Microphone, Scanner, Barcode Reader, Light Pen, Joystick). The input data is not in machine language. Therefore, the input unit converts it into the language of the computer (binary code).

2. Output unit

- the processed information are resulted within the Output devices (to screen, to printer, Speakers, to control other devices). The results of processing are always in binary code. Humans cannot understand this code. Therefore the output unit converts it into a form that we can understand.

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1- What is a Computer?

Four units are found in every computer:

3. Memory unit

- It is classified into two parts **Primary memory, Secondary memory)**

1- Primary memory: it is called the main memory of the computer which is present internally. Here, the data and instructions are stored while the processing is taking place in the CPU. RAM (Random Access Memory) is a primary memory which is volatile (data is lost when the power is disconnected). Another primary memory is ROM(Read Only Memory). Their features are Rapid access, low capacity, stores input information.

2- Secondary memory

It is used for storing the data permanently within external storage devices. Their features are Cheap, long-term, high-capacity storage

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1- What is a Computer?

Four units are found in every computer:

4. Central Processing Unit (CPU)

- ▣ it performs all the processing parts of the computer. The CPU carries out and performs different calculations and other operations on the data and instructions.

It has two subparts:

- ▣ **Arithmetic and Logical Unit:** it is responsible for performing arithmetic tasks such as addition, subtraction, multiplication, division and also making logical decisions such as greater than, less than, ... etc. Hence, its name is also the 'brain' of the computer.
- ▣ **Control Unit:** This unit is responsible for looking after all the processing being done. It organizes and manages the execution of tasks that the CPU performs. In addition, it Supervises and coordinates the various components of the computer

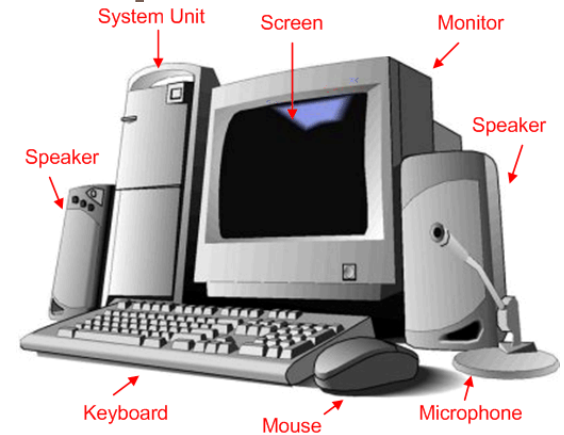
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1- What is a Computer?

Another classification of components for the computer is Hardware and Software

- **Computer Hardware** would include all the tangible parts of a computer.
- **Computer Software** is a intangible parts and it includes the programs that are installed.. Or the instructions for completing tasks.



Some types of system software are:

- **Operating System:** is basically used for interaction between the user and the computer hardware. It controls all the parts of a computer system and manages them, such as Microsoft Windows, Linux, ... etc.
- **Language Processors:** converts the code given by the user (source code) to computer language (machine code).
- **System Utilities:** is responsible for the proper and smooth functioning of computers. And also, keeping the system safe. Such as Antivirus Software, File Management Tools, ... etc.
- **Device Drivers:** basically has the instructions in order to run the hardware devices.
- **Application Software:** performs only a specific task. Such as MS-Excel, PowerPoint, Google Chrome, ... etc.

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2- Why a Computer is?

It is innovated for solving complex problems, fastly it solved most problems in our life

PC at Home

Common uses for the computer within the home

- Computer games
- Working from Home
- Banking from Home
- Connecting to the Web



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2- Why a Computer is?

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Computers in Education

- Computer Based Training (CBT)
- Online learning
- Distance Learning
- Fixed e-learning
- Adaptive e-learning
- Synchronous remote system
- Asynchronous learning
- Blended e-learning system
- Collaborative e-learning
- Massive Online Courses (MOOCs)

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2- Why a Computer is?

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Office Applications

- Automated Production Systems

- Many car factories are almost completely automated and the cars are assembled by computer-controlled robots.
- This automation is becoming increasingly common throughout industry.

- Design Systems

- Many products are designed using CAD (Computer Aided Design) programs to produce exact specifications and detailed drawings on the computer before producing models of new products.

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2- Why a Computer is?

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Office Applications

• Stock Control

- Stock control is ideal for automation and in many companies it is now completely computerized.
- The stock control system keeps track of the number of items in stock and can automatically order replacement items when required.

• Accounts / Payroll

- In most large organizations the accounts are maintained by a computerized system.
- Due to the repetitive nature of accounts a computer system is ideally suited to this task and accuracy is guaranteed.

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2- Why a Computer is?

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Computers in Daily Life

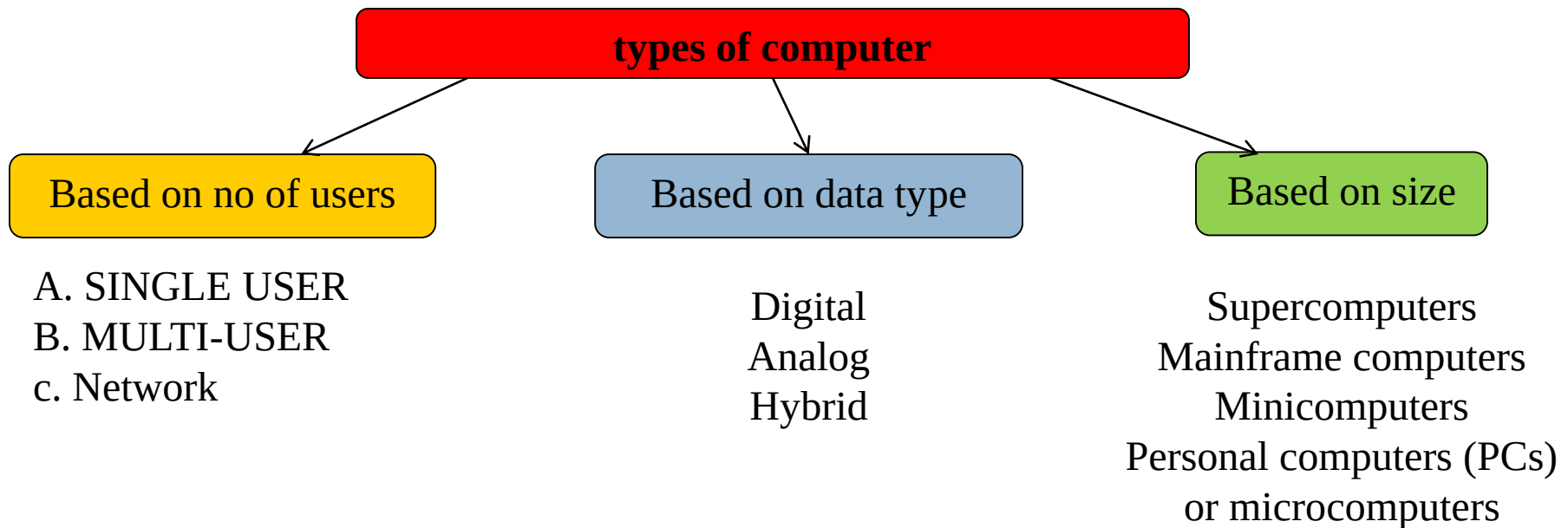
- Accounts
- Games
- Educational
- On-line banking
- Smart ID cards
- Supermarkets
- Working from home (Tele-working)
- Internet



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3- what are types of computer?



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3- what are types of computer?

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Based on size

Supercomputers: have extremely large storage capacity and computing speeds which are many times faster than other computers. More over, it is measured in terms of tens of millions Instructions per second (mips)

Mainframe computers: have storage capacities and very high speed of processing (compared to mini- or microcomputers) are known as mainframe computers. They are multi-programming, high-performance and multi-user computers, which implies they can manage the workload of more than 100 users at a time on the computer.

Minicomputers: are digital and multi-user computer systems with the connection of more than one CPU. Thus, multiple users can work on these computers simultaneously. They are smaller than Mainframe computers and larger than micro computer.

Personal computers (PCs) or Microcomputers: are small computers incorporating a microprocessor, Central Processing Unit (CPU), memory, storage area, an input unit, and an output unit.

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3- what are types of computer?

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Based on data type

Analog computers process an analog data which is continuous data that varies continuously and cannot have discrete values. They measure physical units like the voltage, electric current, pressure, temperature and convert them into digits. Such computers are mostly used for scientific, technology, research, engineering, and industrial applications.

The digital computers execute calculations and logical operations at a high pace. Such computers are proficient in solving problems in discrete formats. It acquires the raw data as input in the form of digits/binary numbers (i.e 0 and 1) and processes it with programs stored in its memory to produce the output.

Hybrid computers can process both continuous and discrete data. It take the features of both Analog and Digital computers. It is fast like an analog computer and has memory and accuracy like those of digital computers. It takes analog signals and transforms them into digital form before processing them.

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3- what are types of computer?

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Based on no of users

- A. SINGLE USER:** Only one user can use the resource at any time.
- B. MULTI-USER:** A single computer is shared by a number of users at any time.
- C. NETWORK:** A number of interconnected autonomous computers shared by a number of users at any time.

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4- what is a Programming Language?

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A set of instructions that tell a computer what to do. In addition, it achieves human-computer interaction

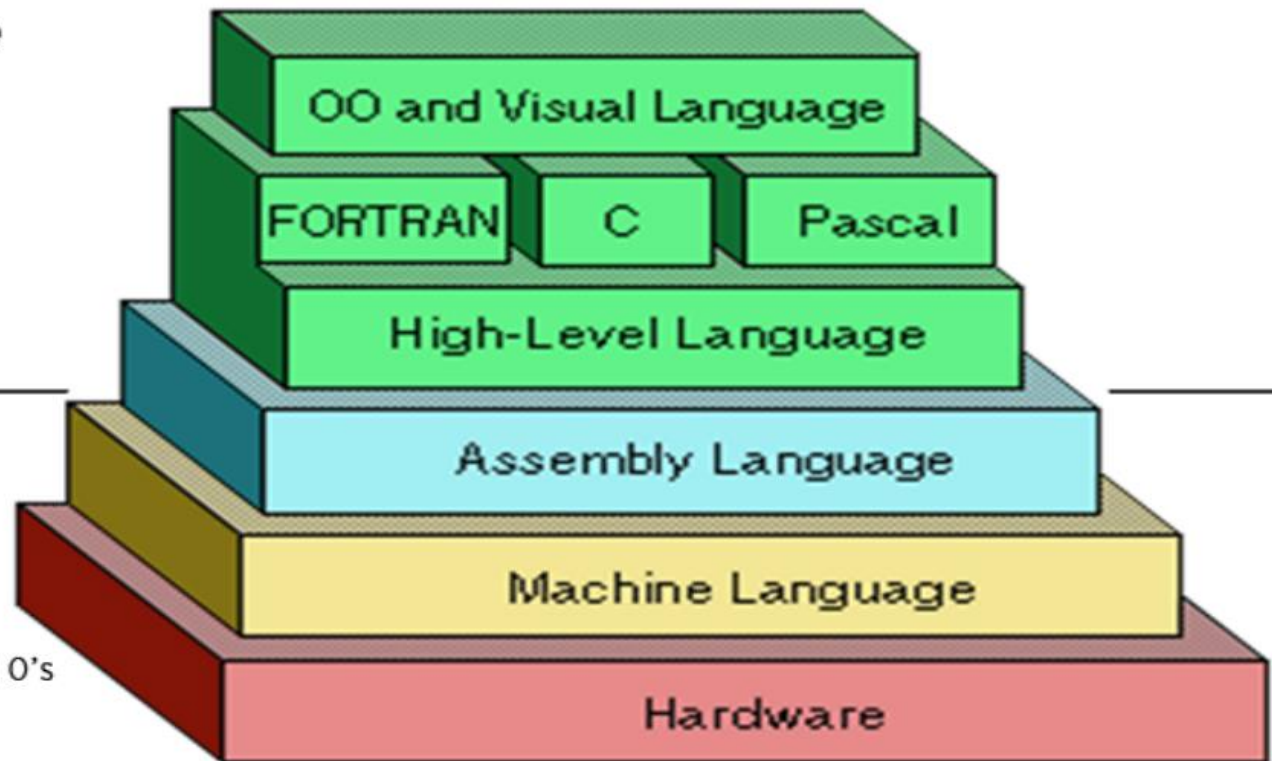


4- what is a Programming Language?

Types of programming languages

High Level Language

- Easy for Programmers to understand
- Contains English Words



Low Level Language

- The computer's own Language
- Binary numbers, in 1's and 0's

Types of programming languages

1- Low Level Language: it includes only 1's and 0's. This language was used in first and second generation computers. In addition, it is very easily understood by a computer but hard to understand for Humans, it is designed to interact with computer hardware. It are categorized into two types: Machine level language and Assembly level language.

Machine level language:

is developed by only using binary numbers i.e., 0 and 1. So, the instructions or the statements in this language use a sequence of 0's and 1's.

Example: 0101000101

Assembly level language:

uses symbols, which are popularly known as mnemonics in computer terminology to write the instructions. So, it is more understandable to the Human than Machine Language.

Example: LOAD a,b , CLR, MOV, PUSH

2- High Level Language: it uses words and commands along with symbols and numbers. The keywords used in High level languages are similar to English words and can be easily understood by Humans when compared to a Low level language. It is classified into two types **Structured and object – oriented programming**

1. Structured Programming:

It divides a program into a set of functions or modules so it is called **Modular programming**. The main function calls the other functions with different parameters. The data is separated from its behavior.

Structured Programming Languages – BASIC, FORTRAN, COBOL, ALGOL, Pascal, and C.

2. Object Oriented Programming:

It depends in programming on the concept of object which contains the data and its behavior. This concept enables programmer to interact with a real-world scenarios. In addition, the class concept which is a blueprint for creating an object. It is necessary to have a class to create.

Object Oriented Programming Languages Java, Python, C#, C++, JavaScript, PHP, Ruby, Dart, Perl, Swift, Scala.

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4- what is a Programming Language?

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Structured Programming

A programming paradigm which divides the code into modules or functions

Focuses on dividing the program into a set of functions in which each function works as a subprogram

Difficult to modify structured programs

Main method communicates with functions by calling those functions in the main program

There are no access specifies

Data is not secured

Difficult to reuse code

Object Oriented Programming

A programming paradigm is based on the concept of objects, which contain data with their behavior

Focuses on representing the program using a set of objects in which encapsulates data and object

Easier to modify object oriented programs

Objects communicate with each other by passing messages

There are access specifies such as private, public and protected

Data is secure

Easy to reuse code

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4- why is a Programming Language?

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a **Programming Language** is used to solve the problem. **The problem** can be identified as facing something hard or difficult to handle. In addition, it is considered the gap between expectations and reality.



Problem solving: is the **process of finding solutions** to difficult or complex issues.

Problem solving is a set of activities designed to **analyze a situation systematically** and generate, implement, and evaluate solutions.

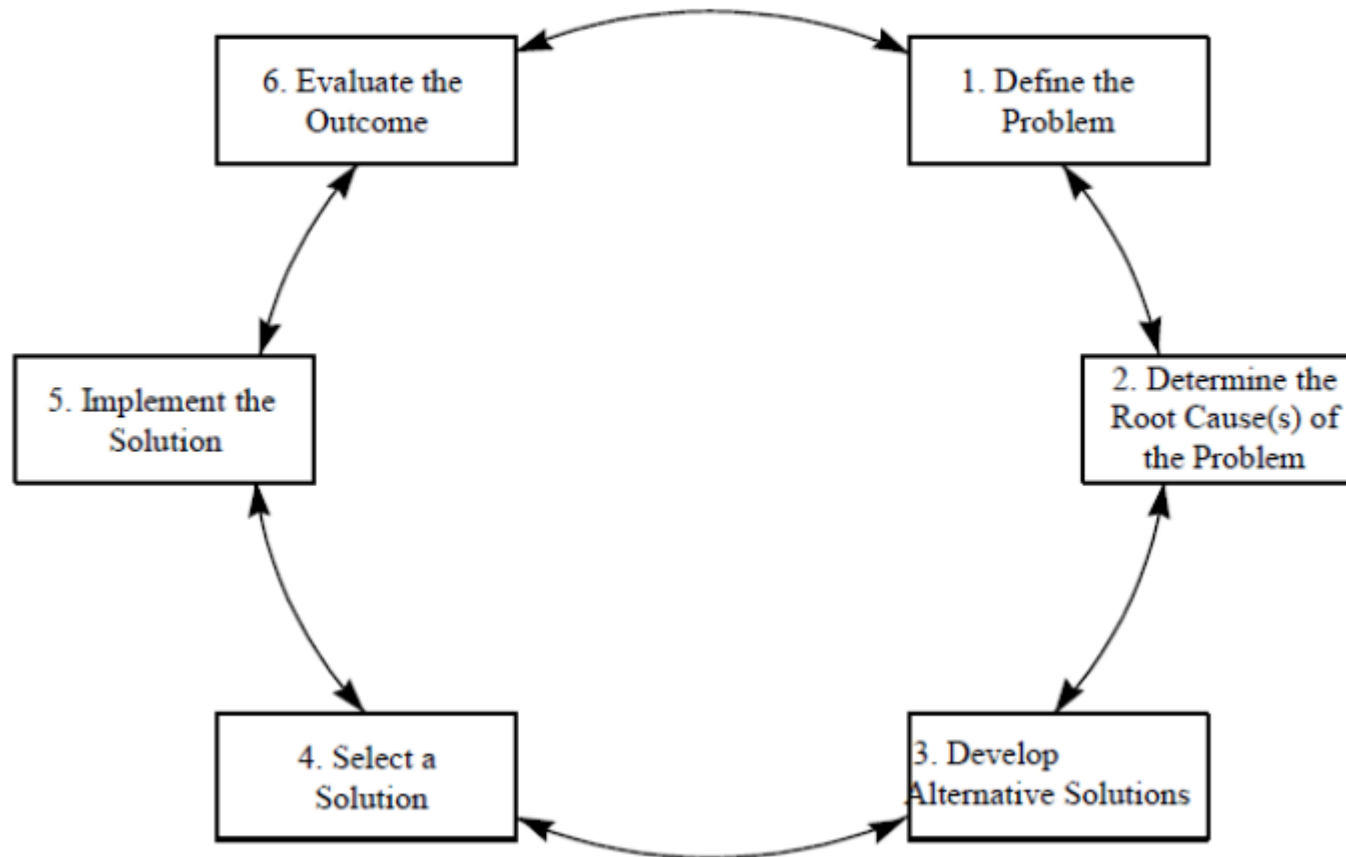


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4- why is a Programming Language?

The Problem Solving Model

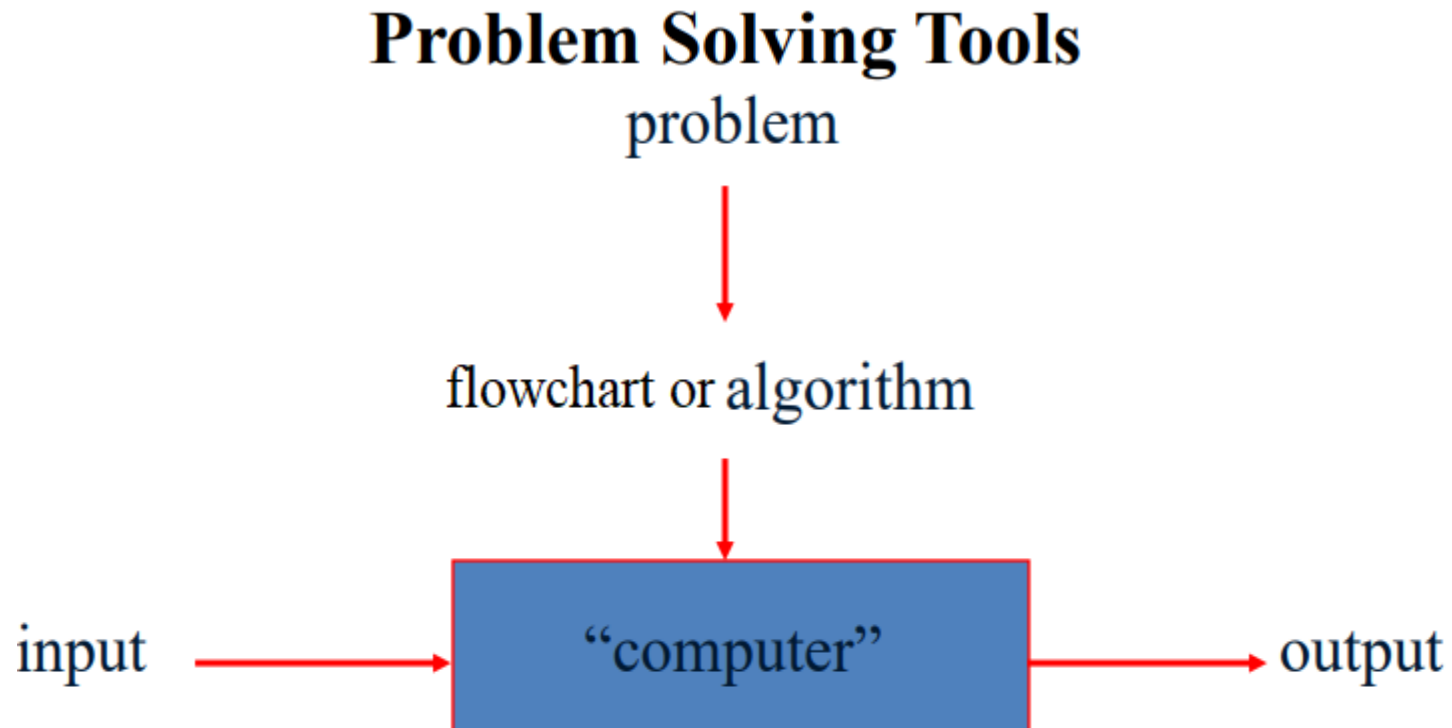


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4- why is a Programming Language?

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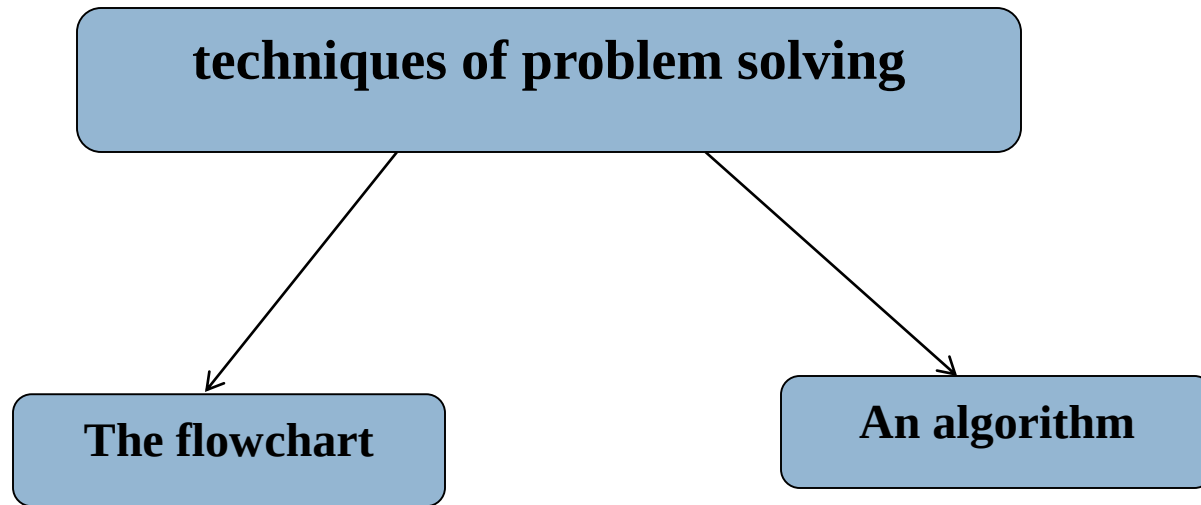


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4- why is a Programming Language?

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is a graphical representation of instructions and decisions to solve a particular problem. it involves a set of boxes and other shapes which are linked or connected by arrows. Each shape represents a step in the process, and the arrows show the order in which they occur. It helps to analyze the problem and plan its solution in a systematic.

is a set of sequential steps usually are written in ordinary language to solve a particular problem. It uses a set of input(s) and produces the desired output.

Properties of an efficient algorithm

- **Finiteness:** An algorithm must terminate in a finite number of steps
- **Definiteness:** Each step of the algorithm must be clear and unambiguous.
- **Effectiveness:** Each step must be effective, in the sense that it should be primitive easily convertible into program statement , can be performed exactly in a finite amount of time.(steps must be doable).
- **Input/output:** Each algorithm must take zero, one or more quantities as input data and produce one or more output values.
- **Generality:** The algorithm must be complete in itself so that it can be used to solve problems of a specific type for any input data.

Advantages of Flowcharts

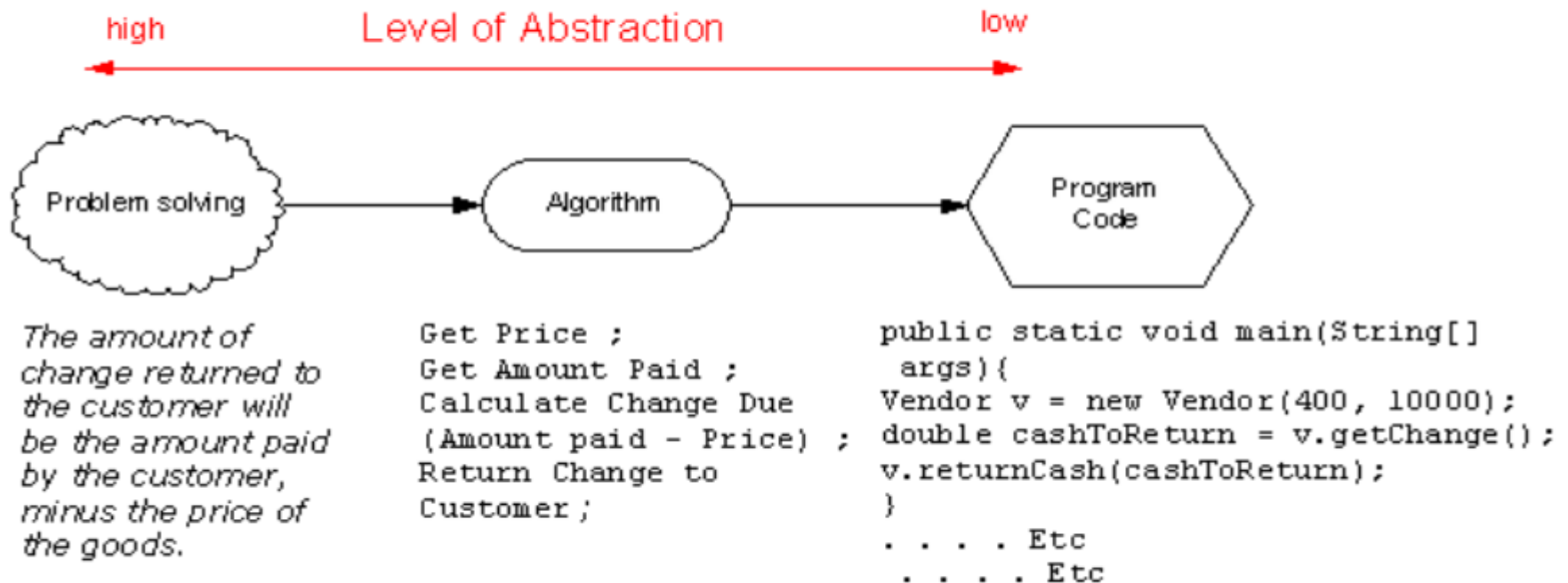
1. The flowchart shows the logic of a problem displayed in pictorial fashion which facilitates **easier checking of an algorithm.**
2. The Flowchart is good means of **communication** to other users. It is also a **compact** means of recording an algorithm solution to a problem.
3. The flowchart allows the problem solver to **break the problem into parts.** These parts can be connected to make **master chart.**
4. The flowchart is a **permanent record** of the solution which can be consulted at a later time.

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4- why is a Programming Language?

- In problem solving, algorithm development and coding we tend to view problems at different **levels of abstraction**



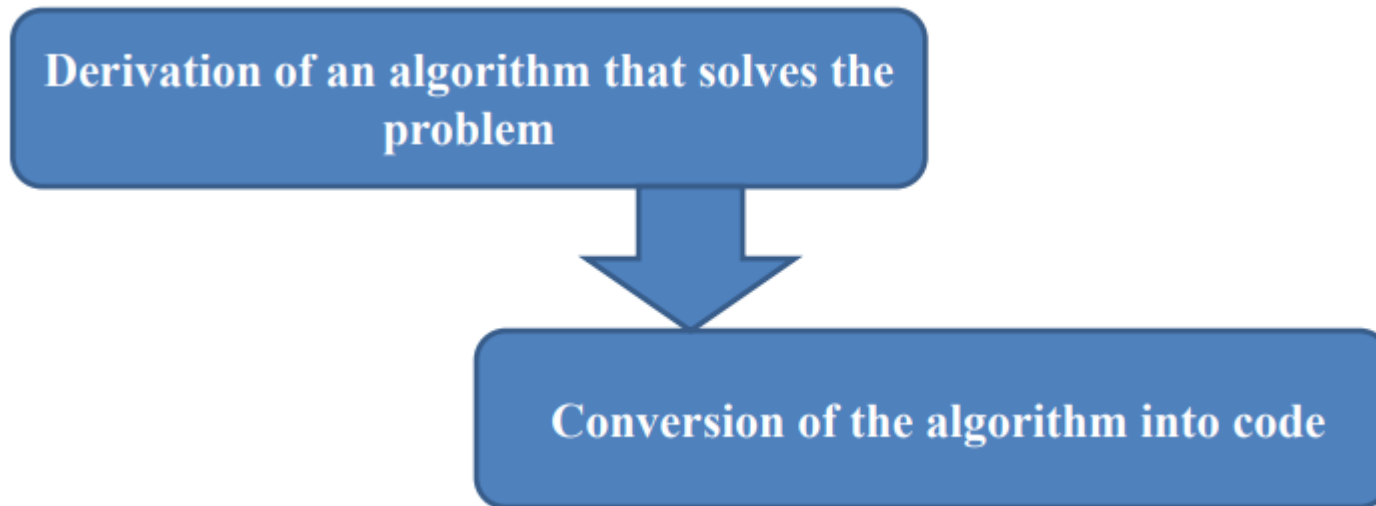
Level of abstraction: Amount of complexity by which a system is viewed
High level means less detail

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4- why is a Programming Language?

- An algorithm, is a form that embeds the **complete logic** of the solution. Its formal written version is called a **computer program**, or **code**.
- Thus, **algorithmic problem solving** actually comes in two phases:



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4- why is a Programming Language?

- Computer programming can be divided into three distinct areas of activity:
- 1. Problem definition and problem solving
- 2. Creating a structured solution (or algorithm)
- 3. Coding (e.g. Java, Python, C++)

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4- why is a Programming Language?

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Three constructs

```
do action 1  
do action 2  
...  
...  
do action  $n$ 
```

a. Sequence

```
if a condition is true,  
then
```

```
do a series of actions
```

```
else
```

```
do another series of actions
```

b. Decision

```
while a condition is true,
```

```
do action 1  
do action 2  
...  
...  
do action  $n$ 
```

c. Repetition

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4- why is a Programming Language?

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Example 1

Find the average of three numbers

- Step 1: Read the numbers a, b, c
- Step 2: Compute the sum of a, b and c
- Step 3: Divide the sum by 3
- Step 4: Store the result in variable d
- Step 5: Print the value of d
- Step 6: End

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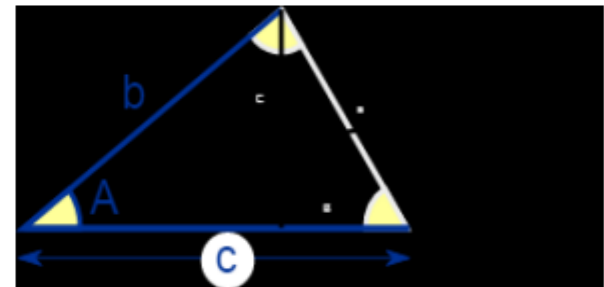
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4- why is a Programming Language?

Example 2

Write an algorithm to find the area of the triangle. Let b, c be the sides of the triangle ABC and A the included angle between the given sides.

- **Step 1:** Input the given elements of the triangle namely sides b, c and angle between the sides A.
- **Step 2:** $\text{Area} = (1/2) * b * c * \sin A$
- **Step 3:** Output the Area
- **Step 4:** End



Example 3

Write an algorithm to compute the sum of the first 100 positive integers.

1. Set sum to 0.
2. Set x to 1.
3. Repeat while $x \leq 100$:
 - Set sum to $\text{sum} + x$.
 - Set x to $x+1$.
4. Print sum.

Example 4

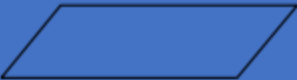
Write an algorithm to find the largest of three numbers a, b, c

- Step 1: Read the numbers a, b, c .
- Step 2: if ($a > b$)
- Big = a
- else BIG = b
- Step 3: if ($BIG < c$)
- Step 4: Big = c
- Step 5: Print the largest number i.e. Big
- Step 6: Stop.

Advantages of Flowcharts

1. The flowchart shows the logic of a problem displayed in pictorial fashion which facilitates **easier checking of an algorithm.**
2. The Flowchart is good means of **communication** to other users. It is also a **compact** means of recording an algorithm solution to a problem.
3. The flowchart allows the problem solver to **break the problem into parts.** These parts can be connected to make **master chart.**
4. The flowchart is a **permanent record** of the solution which can be consulted at a later time.

Symbols used in Flowcharts

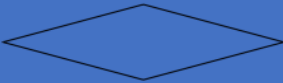
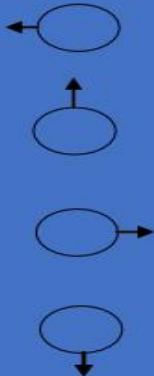
Symbol	Name	Function
	Oval	used to indicate either START/ STOP of the program.
	Input and output indicators	Parallelograms are used to represent input and output operations. Statements like INPUT, READ and PRINT are represented in these Parallelograms.
	Process Indicators	Rectangle is used to indicate any set of processing operation such as for storing arithmetic operations .

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4- why is a Programming Language?

Symbols used in Flowcharts

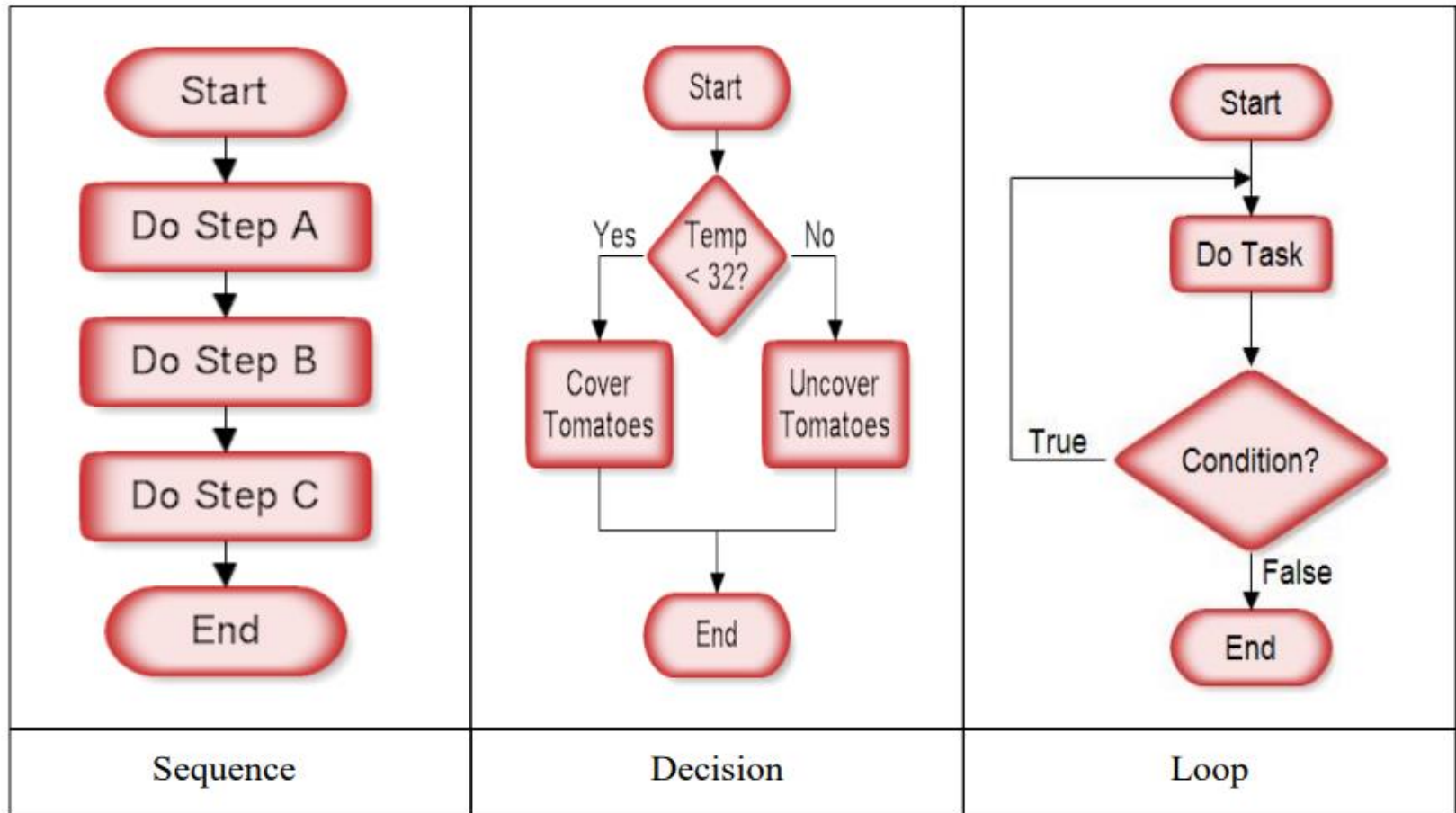
	Decision Makers	The diamond is used for indicating the step of decision making and therefore known as decision box. Decision boxes are used <u>to test the conditions</u> or <u>ask questions</u> and depending upon the answers, the appropriate actions are taken by the computer.
	Flow Lines	Flow lines indicate the direction being followed in the flowchart. In a Flowchart, every line must have an arrow on it to indicate the direction. The arrows maybe in any direction.
	On- Page connectors	Circles are used to join the different parts of a flowchart and these circles are called on-page connectors. The uses of these connectors give a neat shape to the flowcharts. In complicated problems, a flowchart may run in to several pages. The parts of the flowchart on different pages are to be joined with each other. The parts to be joined are indicated by the circle.

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4- why is a Programming Language?

Flowcharts' main structures



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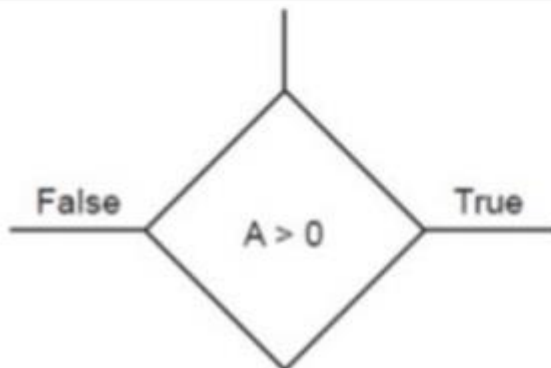
4- why is a Programming Language?

Decision Binary(structure)

- Note:** there are two acceptable ways to represent a decision in all of the structures. Either method is acceptable.

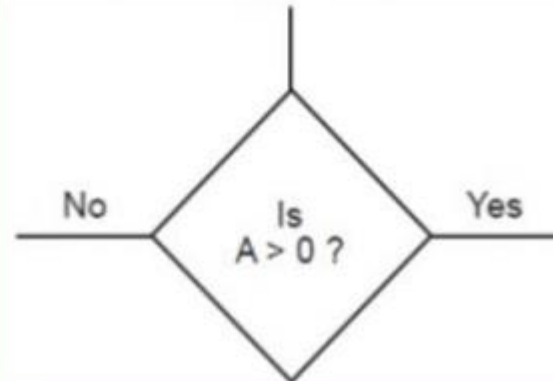
1. The **condition** is expressed as a **statement** and the two possible outcomes are indicated by

- True
- False



2. The **condition** is expressed as a **question** and the two possible outcomes are indicated by

- Yes
- No



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4- why is a Programming Language?

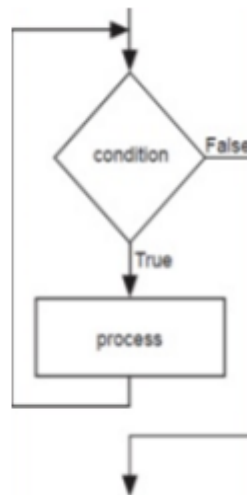
Repetition Structure

An essential feature of repetition is that each loop has a termination condition to stop the repetition.

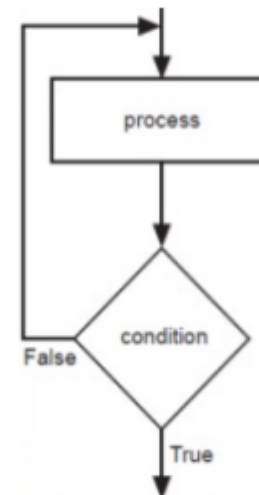
The **termination condition** can be checked or **tested**

1. At the **beginning** and is known as a **pre-test loop** or
2. At the **end** of the loop and is known as a **post-test loop**.

Pre-test Repetition flowchart



Post-test Repetition flowchart



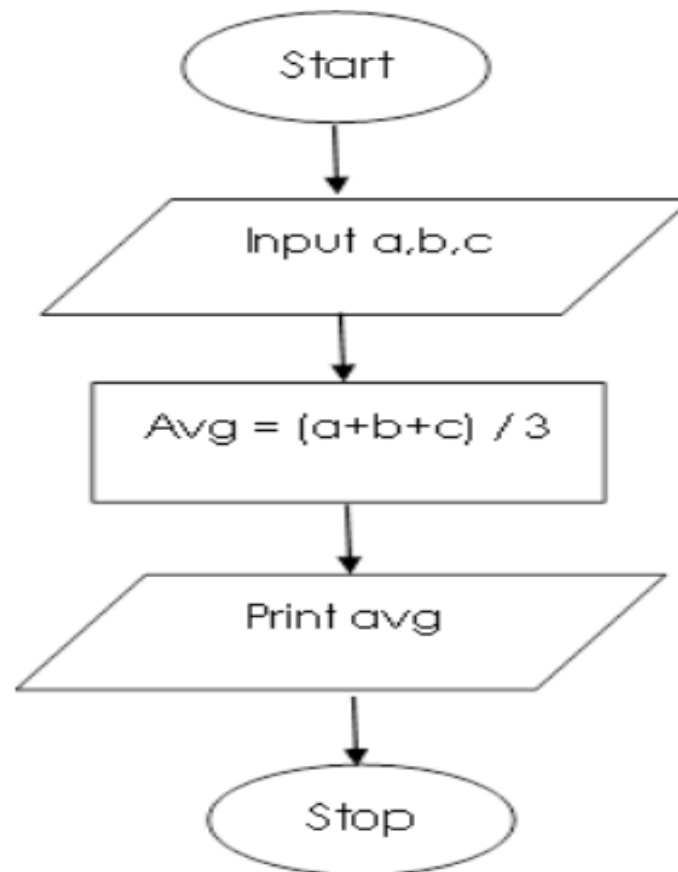
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4- why is a Programming Language?

Example 1

Draw the Flowchart to find the average of three numbers



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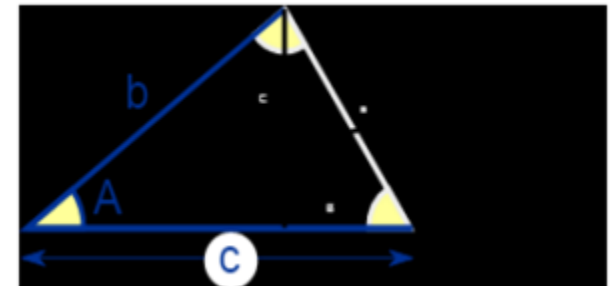
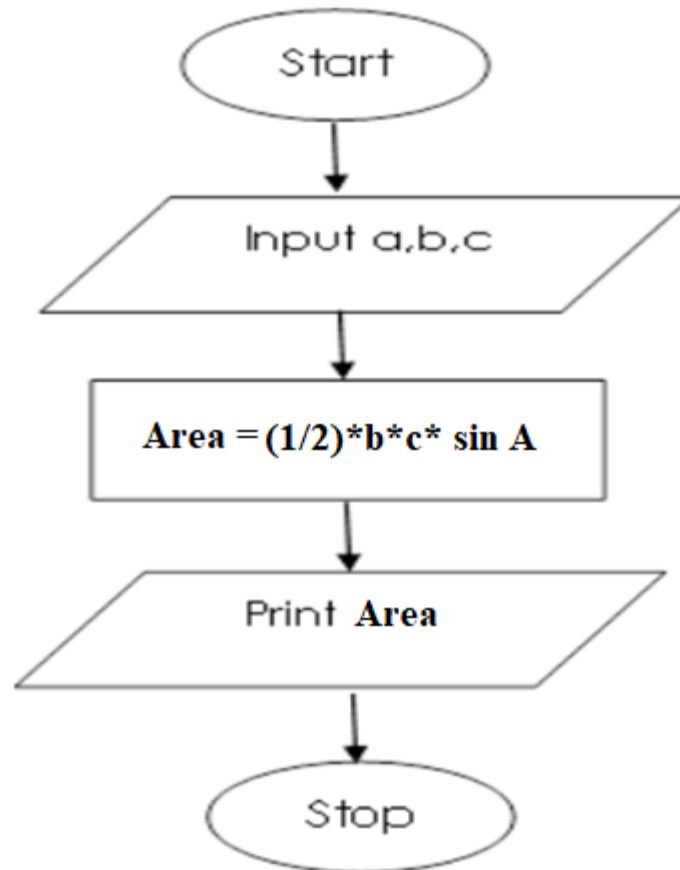
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4- why is a Programming Language?

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Example 2

Write an algorithm to find the area of the triangle. Let b, c be the sides of the triangle ABC and A the included angle between the given sides.



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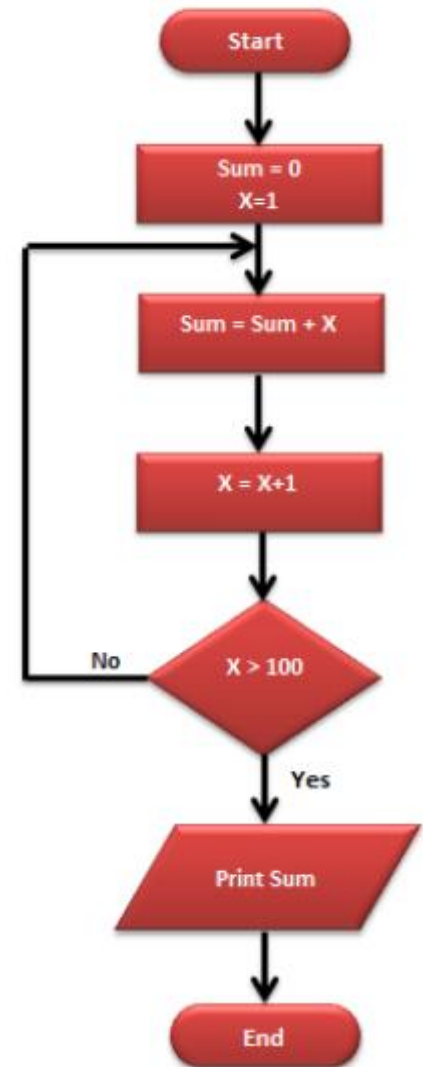
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4- why is a Programming Language?

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Example 3

Draw the chart to compute the sum of the first 100 positive integers.



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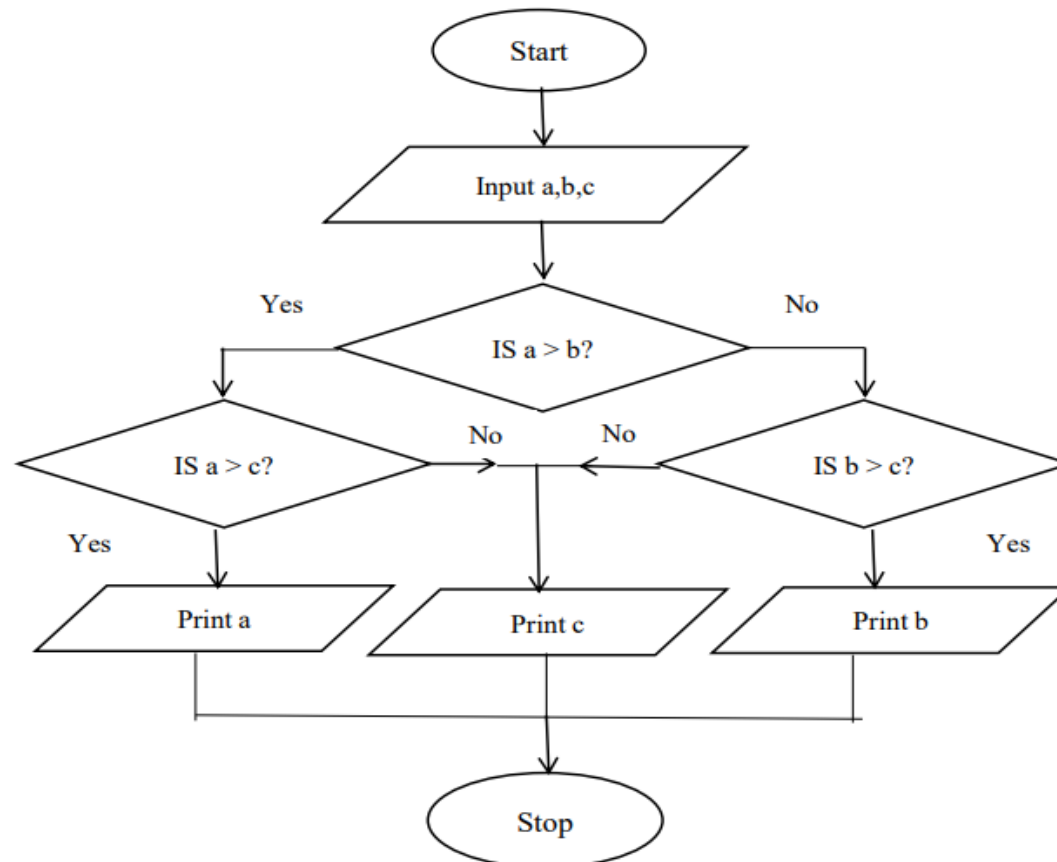
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4- why is a Programming Language?

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Example 4

Write an algorithm to find the largest of three numbers a, b, c



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4- why is a Programming Language?

Differences between Algorithm and Flowchart

Algorithm	Flowchart
A method of representing the step-by-step logical procedure for solving a problem.	A diagrammatic representation of an algorithm. It is constructed using different types of boxes and symbols.
It contains step-by-step English descriptions, each step representing a particular operation leading to solution of problem.	The flowchart employs a series of blocks and arrows, each of which represents a particular step in an algorithm.
These are particularly useful for small problems.	These are useful for detailed representations of complicated programs.
For complex programs, algorithms prove to be Inadequate.	For complex programs, Flowcharts prove to be adequate.

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4- why is a Programming Language?

Example

Find the average of three numbers using an algorithm and flowchart as solving techniques

The solution by an algorithm

Step 1: Read the numbers a, b, c

Step 2: Compute the sum of a, b and c

Step 3: Divide the sum by 3

Step 4: Store the result in variable d

Step 5: Print the value of d

Step 6: End

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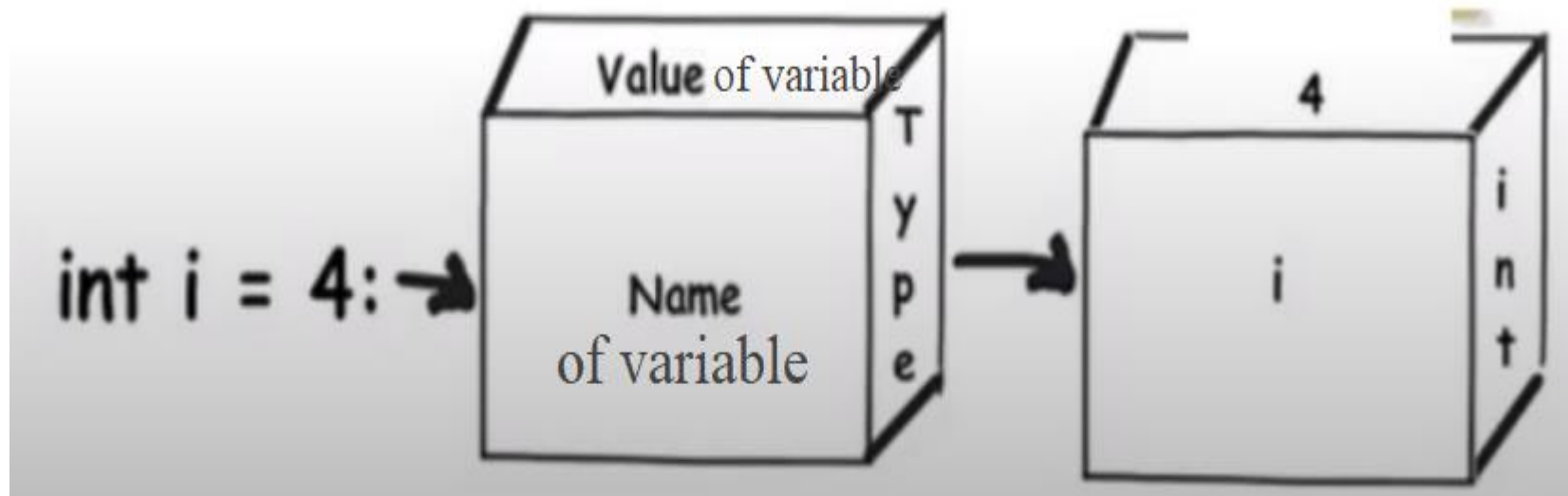
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4- why is a Programming Language?

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What is variable?

It is defined as reservation of place in memory to store specific value with specific data type. This mean that is container of data as following figure:



how the variable is stored in the memory?

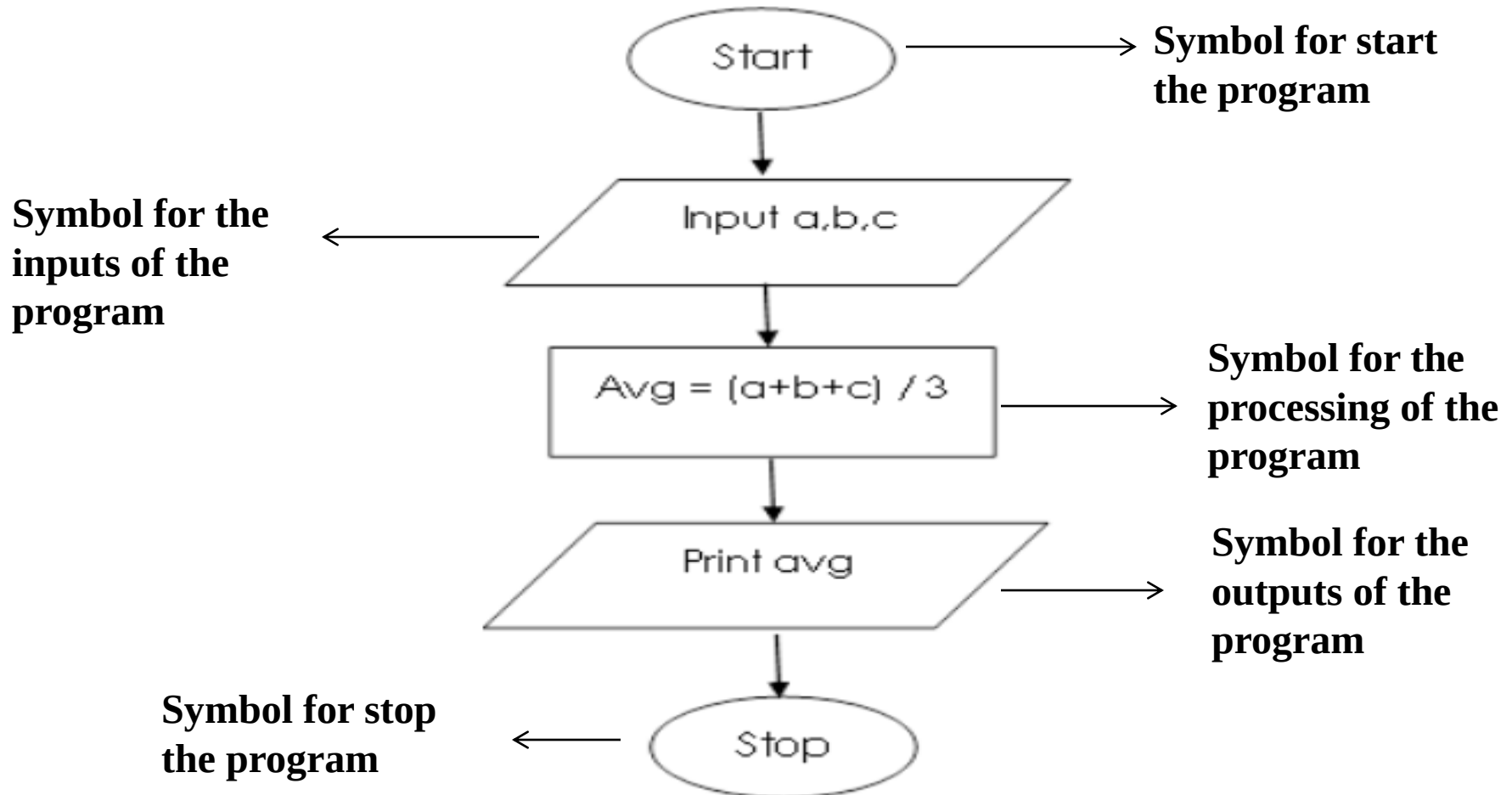
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4- why is a Programming Language?

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The solution by a flowchart



What is C++ Programming Language?

